Expanding the business of Adventure Works

Methodology

For the second business problem, we were presented with the task of selecting two cities within the USA to open the first ever physical AW Stores. As this is most likely only the initial move of this new business model and AW will continue its physical expansion afterwards, we created the "NewStores" Schema, in order maintain all the data and analysis pertinent to this specific project.

To fully deliver on this task, we had to answer the following:

- **Identify "Off-Limit" Cities**: to try to avoid harming the partners that were most responsible for the success of AW in recent time, the new stores cannot be in a city where a TOP 30 Customer is located.
- **Identify "Best Potential" Cities**: Identify the best cities that correspond to a business defined criteria.

Off-Limit Cities:

While a company might have numerous ways of determining a value of a customer (e.g. time as partners, flexibility in negotiation, credit and so on), we decided on one indicator that is usually part of any great business decision: Revenue. In order to identify who the TOP 30 Customers are, we will analyze the total revenue customers made by selling AW products for the available data and rank them. If one of such customers has more than one store, all stores should be considered Off-Limits.

Through our initial exploratory analysis of the available data, we identified that all revenue data could be found at the Table Sales.SalesOrderHeader (SOH), and the most important customers info (for this case) was already combined in the View Sales.vStoreWithAddresses (vSWA), making this our initial set on to where additional data would be integrated.

Since none of these two sets had any Primary Key/Foreign Key in common, an additional set had to be brought to combine them. The ideal option was the Table Sales.Customer (CS), as it connects to vSWA through StoreID and to the SOH through the CustomerID.

We chose to select the customer Names, their Cities, States and Postal Codes, as well as the sum of their total revenue made through AW products, making sure only to select customers in the USA (Script 01, Table 01).

Afterwards, we checked how many customers presented more than one store, to understand how much we would need to worry about this in the next step. Through vSWA, we were able to count how many different addresses each customer had. This returned that only "Friendly Bike Shop" and "Sports Products Store" had more than one store, each with two addresses (Script 02). While "SPS" not only

did not make the cut to the TOP 30, one of the stores is in France. As for "FBS", the Bellingham store was ranked in the 25th place, meaning both Bellingham and Port Huron will be considered Off-Limits.

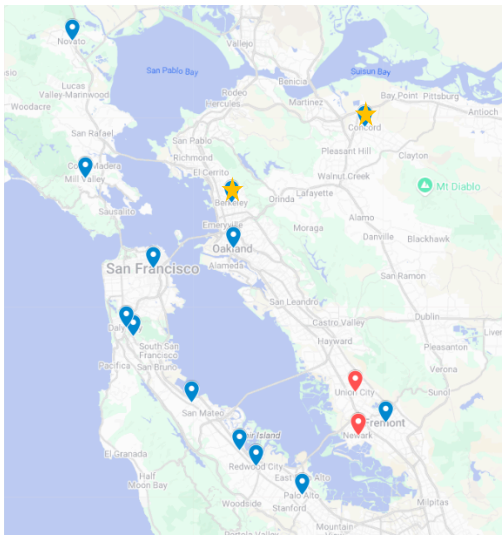
Best Potential Cities:

In similar ways as before, a company might have different criterion on why a city could be considered the best location (brand development, climate affinity, etc.). Thankfully, AW already has a considerable database of where people like and consume their products the most: their individuals sales database. We assumed that the individuals' sales cities were a good proxy to understand which cities already had a good direct relation with the brand, despite being serviced by other stores or not.

Our initial approach was to tackle this in the same manner as in the previous problem. However, no rankings were necessary and there was no need to understand the revenue by individuals, but by the cities where they live. For this instance, no intermediate set was necessary since SOH could be directly linked to the View Sales.vIndividualCustomer (vIC) through the CustomerID. In addition to the City, Postal Code, State and Revenue, we added the Quantity of Sales and the calculated the average Ticket for a better overview (Script 03, Table 02).

We also analyzed which were the best states, following the same line of thought as before. Given the sheer difference, the only states to be considered were California, Washington and Oregon (Script 04, Table 03).

Taking in consideration the similarity of the postal codes from the top two cities, we decided to investigate the distance between them (as well as all other cities with the same initial two digits in the postal code – let us name this 'Postal2' - as well as the off-limit cities in the area).



The result was that while there are two off limit cities in the Bay Area (marked in red), there are several other cities that are home to individual clients (marked in blue), and two of those are the top two cities in the USA for individual clients (marked in yellow stars).

Given the proximity of the cities, we believe it would not make sense to open the stores in Concord and Berkeley, as one well-placed store could be able to impact the region. As such, to understand not only the best possible cities but also the best possible regions and avoid placing a store to close to another, we will also analyze the sales that happen within the Postal2 of a city.

With all of this in mind, we combined the data from which were the best cities by individuals' sales with the total Individual Market and Store Market of their Postal2. The Individual Market brings a

potential revenue if all online sales would convert to the new store. The Store Market brings how well developed the stores are in the area, bringing light to possible cannibalization issues and difficulties to enter the market. In addition to that, we created the 'MarketAvailability' criteria, the share of Individual Market over the total Postal2 Market (the closer to 1 the better, Scripts 05, 06 and 07, Table 04).

Results and Discussion

We are now able to drill down to the following top 5 Postal2 Individual Markets (Script 08, table 05):

- 98, Seattle region, WA: greatest Individual Market, even greater Stores Market, lowest Market Availability, Bellingham being the only city to appear in both Top30 clients and TopIndividual clients lists.
- 94, Bay Area, CA: Greatest Individual Market in California, only market where Individuals Market is greater than Stores Market.
- 91, San Diego region, CA: similar Individual Market to the Bay Area, better developed Stores Market.
- 97, Portland region, OR: similar Market Availability to San Diego region despite the lower Individual Market size, no Top30 clients in the region.
- 90, Los Angeles region, CA: second lowest market availability

Conclusions

Given the five options above, we recommend that the best cities for the new stores are at the Bay Area (locating the store in the largest city, San Francisco) and the Portland region (locating the store in the largest city, Portland). Despite having the highest available volume, the Seattle region would present to much of a difficulty to enter the market, and given the lack of experience with this kind of operation would not be a good idea. As for San Diego, we believe there is a greater advantage to the company to open stores in two distinct markets, and the Bay Area is a clear better choice for California.

Appendix

```sql
IF OBJECT_ID('NewStores.OffLimitCities', 'U') IS NOT NULL
BEGIN
    DROP TABLE NewStores.OffLimitCities;
END

CREATE TABLE NewStores.OffLimitCities (
    StoreName NVARCHAR(100),
    City NVARCHAR(100),
    US_State NVARCHAR(100),
    PostalCode NVARCHAR(20),
    Stores_Revenue DECIMAL(18,2)
);


INSERT INTO NewStores.OffLimitCities (StoreName, City, US_State, PostalCode, Stores_Revenue)
SELECT TOP 30
    vSWA.Name as StoreName,
    vSWA.City,
    vSWA.StateProvinceName AS US_State,
    vSWA.PostalCode,
    SUM(SOH.TotalDue) AS Stores_Revenue
FROM Sales.vStoreWithAddresses AS vSWA
JOIN Sales.Customer AS CS
    ON vSWA.BusinessEntityID = CS.StoreID
JOIN Sales.SalesOrderHeader AS SOH
    ON CS.CustomerID = SOH.CustomerID
WHERE vSWA.CountryRegionName = 'United States'
GROUP BY
    vSWA.BusinessEntityID,
    vSWA.Name,
    vSWA.City,
    vSWA.StateProvinceName,
    vSWA.PostalCode,
    vSWA.CountryRegionName
ORDER BY SUM(SOH.TotalDue) DESC;
```

Script 01

	StoreName	City	US_State	PostalCode	Stores_Revenue
1	Modular Cycle Systems	Austin	Texas	78701	1321291.88
2	Safe Cycles Shop	Bellevue	Washington	98004	999630.63
3	Brakes and Gears	Tooele	Utah	84074	989184.08
4	Tread Industries	Orlando	Florida	32804	973875.19
5	Excellent Riding Sup…	Memphis	Tennessee	38103	961675.86
6	Hardware Components	Minneap…	Minnesota	55402	920349.72
7	Totes & Baskets Comp…	San Ant…	Texas	78204	919801.82
8	Outdoor Equipment St…	Nashua	New Hampsh…	03064	841866.55
9	Thorough Parts and R…	Lacey	Washington	98503	834475.93
10	Fitness Toy Store	Garland	Texas	75040	820383.55
11	Latest Sports Equipm…	Seattle	Washington	98104	815914.25
12	First Bike Store	Odessa	Missouri	64076	801766.21
13	Great Bikes	Casper	Wyoming	82601	789884.29
14	Farthermost Bike Shop	Carson	California	90746	781073.72
15	Field Trip Store	Loveland	Colorado	80537	757449.68
16	Eastside Department …	Union C…	California	94587	718258.81
17	The Gear Store	Houston	Texas	77003	696703.61
18	Sheet Metal Manufact…	College…	Texas	77840	695242.23
19	Golf and Cycle Store	La Mesa	California	91941	665292.38
20	Permanent Finish Pro…	Reno	Nevada	89502	650362.05
21	Reasonable Bicycle S…	Greeley	Colorado	80631	644168.16
22	Westside Plaza	Sand Ci…	California	93955	606819.61
23	Larger Cycle Shop	Melville	New York	11747	605358.54
24	Closeout Boutique	Seattle	Washington	98104	604794.78
25	Friendly Bike Shop	Belling…	Washington	98225	600077.27
26	Riders Company	Branch	Minnesota	55056	584949.13
27	Resale Services	Culver …	California	90232	554388.94
28	Mail-Order Outlet	Ellensb…	Washington	98926	538333.47
29	Mountain Bike Center	Newark	California	94560	514317.53
30	eCommerce Bikes	Gulfport	Mississippi	39501	502367.04

Table 01

```sql
-- Identify Clients with more than one store:
select
    Name
    , COUNT(Name)
from Sales.Store
group by Name
having COUNT(Name) > 1

---- Two Clients were identified:
    -- Friendly Bike Shop
    -- Sports Products Store

---- Cities where those two clients are:
Select * from Sales.vStoreWithAddresses
where Name = 'Friendly Bike Shop' OR Name = 'Sports Products Store'
    -- Bellingham/US (TOP25), Port Huron/US
    -- Lieusaint/FR, Santa Ana/US
```

Script 02

```sql
-- Best City Analysis:
select
    vIC.City,
    vIC.PostalCode,
    vIC.StateProvinceName,
    SUM(SOH.TotalDue) as Individuals_Revenue,
    COUNT(SOH.SalesOrderID) as QuantityOfSales,
    SUM(SOH.TotalDue)/COUNT(SOH.SalesOrderID) as Ticket
from Sales.vIndividualCustomer as vIC
join Sales.SalesOrderHeader as SOH
on vIC.BusinessEntityID = SOH.CustomerID
where vIC.CountryRegionName = 'United States'
group by vIC.City, vIC.PostalCode, vIC.StateProvinceName
order by SUM(SOH.TotalDue) DESC
```

Script 03

	City	PostalCode	StateProvinceName	Individuals_Revenue	QuantityOfSales	Ticket
1	Concord	94519	California	311348,6775	200	1556,7433
2	Berkeley	94704	California	270041,3589	198	1363,8452
3	Beverly Hills	90210	California	264573,781	202	1309,7711
4	Beaverton	97005	Oregon	258131,7444	201	1284,2375
5	Bellingham	98225	Washington	235024,8893	242	971,1772
6	Chula Vista	91910	California	234170,2847	214	1094,2536
7	Burbank	91502	California	231956,526	163	1423,0461
8	Burien	98168	Washington	228318,0451	214	1066,9067
9	Burlingame	94010	California	218126,4342	211	1033,7745
10	Bremerton	98312	Washington	217370,4499	155	1402,3899
11	Bellflower	90706	California	183000,8969	181	1011,0546
12	Colma	94014	California	182889,1462	179	1021,727
13	Coronado	92118	California	171831,6113	149	1153,2322
14	Yakima	98901	Washington	164729,7719	88	1871,9292
15	Corvallis	97330	Oregon	157341,0262	97	1622,0724

Table 02

```sql
-- Best States Analysis:
select
    vIC.StateProvinceName,
    SUM(SOH.TotalDue) as Individuals_Revenue
from Sales.vIndividualCustomer as vIC
join Sales.SalesOrderHeader as SOH
on vIC.BusinessEntityID = SOH.CustomerID
where vIC.CountryRegionName = 'United States'
group by vIC.StateProvinceName
order by SUM(SOH.TotalDue) DESC
```

Script 04

	StateProvinceName	Individuals_Revenue
1	California	5299957,2692
2	Washington	2583613,2536
3	Oregon	1179621,5227
4	Texas	11974,7746
5	Illinois	10433,2646
6	Wyoming	9985,0923
7	Arizona	9652,189
8	Mississippi	9178,4616
9	Utah	7357,731
10	Virginia	7344,471
11	Ohio	7267,3641
12	Florida	3855,1383
13	New York	3314,9002
14	Montana	1303,3144
15	Georgia	254,4263
16	Minnesota	101,8369

Table 03

```sql
IF OBJECT_ID('NewStores.MarketIndividuals', 'U') IS NOT NULL
BEGIN
    DROP TABLE NewStores.MarketIndividuals;
END

CREATE TABLE NewStores.MarketIndividuals (
    Postal NVARCHAR(10),
    Postal_Individual_Revenue DECIMAL(18,2)
);


INSERT INTO NewStores.MarketIndividuals (Postal, Postal_Individual_Revenue)

select LEFT(vIC.PostalCode, 2) as Postal,
    SUM(SOH.TotalDue) as Postal_Individual_Revenue
from Sales.vIndividualCustomer as vIC
join Sales.SalesOrderHeader as SOH
on vIC.BusinessEntityID = SOH.CustomerID
where vIC.CountryRegionName = 'United States'
group by LEFT(vIC.PostalCode, 2)
order by SUM(SOH.TotalDue) DESC

select * from NewStores.MarketIndividuals
order by Postal_Individual_Revenue DESC
```

Script 05

```sql
IF OBJECT_ID('NewStores.MarketStores', 'U') IS NOT NULL
BEGIN
    DROP TABLE NewStores.MarketStores;
END

CREATE TABLE NewStores.MarketStores (
    Postal NVARCHAR(10),
    Postal_Stores_Revenue DECIMAL(18,2)
);


INSERT INTO NewStores.MarketStores (Postal, Postal_Stores_Revenue)

SELECT
    LEFT(vSWA.PostalCode, 2) AS Postal,
    SUM(SOH.TotalDue) AS Postal_Stores_Revenue
FROM Sales.vStoreWithAddresses AS vSWA
JOIN Sales.Customer AS CS
    ON vSWA.BusinessEntityID = CS.StoreID
JOIN Sales.SalesOrderHeader AS SOH
    ON CS.CustomerID = SOH.CustomerID
WHERE vSWA.CountryRegionName = 'United States'
GROUP BY
    LEFT(vSWA.PostalCode, 2)
ORDER BY SUM(SOH.TotalDue) DESC

select * from NewStores.MarketStores
order by Postal_Stores_Revenue DESC
```

Script 06

```sql
IF OBJECT_ID('NewStores.BestCities', 'U') IS NOT NULL
BEGIN
    DROP TABLE NewStores.BestCities;
END

CREATE TABLE NewStores.BestCities (
    City NVARCHAR(100),
    PostalCode NVARCHAR(10),
    Postal2 INT,
    StateProvinceName NVARCHAR (100),
    Individuals_Revenue DECIMAL (18,2),
    QuantityOfSales INT,
    Ticket DECIMAL (18,2),
    OffLimitCities NVARCHAR (100)
);


INSERT INTO NewStores.BestCities (City, PostalCode, Postal2, StateProvinceName, Individuals_Revenue, QuantityOfSales, Ticket, OffLimitCities)


select vIC.City,
    vIC.PostalCode,
    LEFT(vIC.PostalCode, 2),
    vIC.StateProvinceName,
    SUM(SOH.TotalDue) as Individuals_Revenue,
    COUNT(SOH.SalesOrderID) as QuantityOfSales,
    SUM(SOH.TotalDue)/COUNT(SOH.SalesOrderID) as Ticket,
    CASE
        WHEN OLC.City IS NULL THEN 'OK'
        ELSE 'Off Limit'
        END AS OffLimitCities
from Sales.vIndividualCustomer as vIC
join Sales.SalesOrderHeader as SOH
on vIC.BusinessEntityID = SOH.CustomerID
left join NewStores.OffLimitCities as OLC
on vIC.City = OLC.City
where vIC.CountryRegionName = 'United States'
group by vIC.City, vIC.PostalCode, vIC.StateProvinceName, OLC.City
order by SUM(SOH.TotalDue) DESC
```

Script 07

	City	PostalCode	Postal2	StateProvi…	Individua…	Quan…	Ticket	OffLimitCities
1	Concord	94519	94	California	311348.68	200	1556.74	OK
2	Berkeley	94704	94	California	270041.36	198	1363.85	OK
3	Beverly Hills	90210	90	California	264573.78	202	1309.77	OK
4	Beaverton	97005	97	Oregon	258131.74	201	1284.24	OK
5	Bellingham	98225	98	Washington	235024.89	242	971.18	Off Limit
6	Chula Vista	91910	91	California	234170.28	214	1094.25	OK
7	Burbank	91502	91	California	231956.53	163	1423.05	OK
8	Burien	98168	98	Washington	228318.05	214	1066.91	OK
9	Burlingame	94010	94	California	218126.43	211	1033.77	OK
10	Bremerton	98312	98	Washington	217370.45	155	1402.39	OK
11	Bellflower	90706	90	California	183000.90	181	1011.05	OK
12	Colma	94014	94	California	182889.15	179	1021.73	OK
13	Coronado	92118	92	California	171831.61	149	1153.23	OK
14	Yakima	98901	98	Washington	164729.77	88	1871.93	OK
15	Corvallis	97330	97	Oregon	157341.03	97	1622.07	OK
16	Seattle	98104	98	Washington	152239.54	134	1136.12	Off Limit
17	El Cajon	92020	92	California	146621.61	100	1466.22	OK
18	Lemon Grove	91945	91	California	146497.93	117	1252.12	OK

Table 04

```sql
SELECT
    Distinct BC.Postal2,
    1-MS.Postal_Stores_Revenue/(MI.Postal_Individual_Revenue+MS.Postal_Stores_Revenue) as MarketAvailability,
    MS.Postal_Stores_Revenue,
    MI.Postal_Individual_Revenue
from NewStores.BestCities as BC
join NewStores.MarketStores as MS
ON BC.Postal2 = MS.Postal
join NewStores.MarketIndividuals as MI
ON LEFT(BC.PostalCode, 2) = MI.Postal
order by MI.Postal_Individual_Revenue DESC
```

Script 08

	Postal2	MarketAvailability	Postal_Stores_Revenue	Postal_Individual_Revenue
1	98	0.22867934413472989245	8145186.52	2414865.85
2	94	0.53455663573577653951	1555753.47	1786765.92
3	91	0.42650857640509740734	2219219.38	1650445.08
4	97	0.40837463187062551024	1708955.36	1179621.52
5	90	0.25865622574415299577	3099989.76	1081592.21
6	92	0.29649133188053253630	1558084.25	656649.30

Table 04